Scheduler Rankings and Performance Trade-offs in a Common-Model LLM-Serving Evaluation

SOROUSH VAHIDI, New Jersey Institute of Technology, USA

A scheduler comparison is difficult to interpret when its implementations use different service, resource, or information rules. We evaluate four project-maintained LLM-serving scheduler adaptations under a shared, synthetic execution contract. The study covers 120 trace-derived windows from three sources, six inherited arrival/deadline settings, and two KV reservation capacities: 5,760 validated cells, all completing every request. Against chunked-FCFS, the SLO-gated adaptation reduces the six source/capacity aggregate mean latencies by 11.3–34.0%, while reducing token throughput by 1.65–4.01%. On BurstGPT, its mean window-level p95 latency increases by 3.72% and 18.97% at the two capacities. Its urgency gate does not bind; these results concern the adaptation's admission ordering and chunking, not the complete original SLAI algorithm. Chunked-FCFS and decode-first produce identical records in all 1,440 matched conditions. No source-mean comparison exhibits a strict 10% practical reversal, although window-level and metric-dependent reversals occur. Exact ties and redundant rankings further limit what a leaderboard conveys. A separate 134,000-study known-truth calibration diagnostic identifies failures of historical inferential procedures under specified synthetic scenarios. The contribution is a finite-corpus performance evaluation that distinguishes measured trade-offs, model-implied invariances, and inferential limits; it is neither a new scheduler nor a validated native-GPU performance predictor.

CCS Concepts: • **Computing methodologies** → **Simulation evaluation**; • **Computer systems organization** → *Cloud computing*.

Additional Key Words and Phrases: LLM serving, performance evaluation, scheduling, trace-driven simulation, ranking stability

1 Introduction

A serving scheduler can reduce average response time while making long requests slower or lowering aggregate throughput. Which scheduler appears best therefore depends on both the workload and the objective. There is a more basic difficulty in simulation-based comparisons: a policy label can select not only an ordering rule, but also a different prefill model, memory constraint, or information interface. An observed rank difference then combines scheduling decisions with differences in what the evaluator allows those decisions to accomplish.

This problem arises in the historical LLM-Serving Scheduler Portability (LSSP) corpus that motivates our study. Its implementations shared request inputs, but did not uniformly enforce prefill work and KV capacity. Some comparisons therefore concerned policy–model combinations rather than scheduling alternatives within a common execution model. Repeating those runs or reproducing their ranking arithmetic cannot resolve that ambiguity. A useful comparison must first identify the execution contract: which requests are visible, which actions are legal, how work advances, and when resources are acquired and released.

We evaluate four adaptations under such a contract, CM-0.1. It is a CPU discrete-event model with common batch timing, token accounting, information restrictions, and conservative full-context KV reservations. The adaptations retain selected ideas from FCFS serving, chunked prefill, decode-first scheduling, and SLAI-inspired prompt ordering [2, 4, 10]. They have new implementation identities

Author's Contact Information: Soroush Vahidi, sv96@njit.edu, New Jersey Institute of Technology, Newark, New Jersey, USA.

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and a shared action space. The purpose is to measure their behavior under a controlled abstraction, not to reproduce the performance of their published systems.

Workload-sensitive serving evaluations already establish that benchmark choice matters [13]; predictive serving simulators address the cost of exploring configurations [1, 7]. Our contribution is a different empirical object: a complete comparison of restricted adaptations across a fixed corpus, with absolute performance, matched differences, ties, and practical reversals reported together. This permits a reader to distinguish a useful performance difference from a rank change caused by aggregation, duplicate metrics, or an inactive scheduling distinction.

The completed comparison changes the emphasis of the historical narrative. Across six source–capacity aggregates, SLO-gated improves mean latency but reduces throughput; BurstGPT also has worse mean window-p95 latency. Chunked-FCFS and decode-first are exactly equivalent in every matched condition. Source-mean comparisons have no strict 10% practical reversals, whereas individual windows and different metrics do. These findings make an unconditional "best scheduler" an inadequate summary, without implying that rankings are unstable at every aggregation level.

We ask three questions:

RQ1: Performance and operating conditions. How do the absolute performances of four scheduler adaptations vary across workload windows, inherited arrival/deadline settings, and KV reservation capacities?

RQ2: Ranking stability and metric choice. How do workload variation, aggregation, metric selection, ties, and practical performance differences affect scheduler rankings?

RQ3: Interpretation and inference. What do the historical comparison audit and known-truth statistical calibration establish about the reliability and limitations of scheduler comparisons?

The paper makes four specific contributions. First, it specifies an executable comparison contract and identifies the departures of each adaptation from its published lineage. Second, it reports the complete 5,760-cell evaluation, including absolute latency, throughput, goodput, and complete request accounting. Third, it quantifies how window variation, aggregation, ties, and metric choice change the interpretation of rankings, including the null source-mean reversal result. Fourth, it separates execution conformance from statistical calibration through a concise historical audit and a completed synthetic diagnostic. These contributions support finite-corpus conclusions under CM-0.1; they make no new algorithmic, statistical-test, or native-performance claim.

2 Related Work and Evaluation Scope

Serving mechanisms. Orca introduced iteration-level scheduling to change batches as requests progress [18]. vLLM's PagedAttention addresses KV-memory fragmentation and sharing [10]. Sarathi-Serve uses chunked prefill to manage throughput–latency trade-offs [2]. Bari et al. distinguish the throughput-optimal RAD construction from the practical SLAI scheduler, which uses measured decode urgency and known prompt lengths [4]. These are system and algorithm contributions with their own execution assumptions. Our whole-FCFS, chunked-FCFS, decode-first, and SLO-gated adaptations isolate a smaller set of admissible rules. In particular, full-context reservations do not reproduce PagedAttention, and an inactive urgency gate supplies no evidence for SLAI's complete design or RAD's theorem. Disaggregated serving and dynamic migration, exemplified by DistServe and Llumnix [15, 20], are outside the single-device, nonpreemptive contract evaluated here.

Workload-sensitive evaluation. Papaioannou and Doudali explicitly examine how dataset and inference-scenario choice affect comparisons among LLM inference systems [13]. Their work is a close precedent for the importance of workload choice, rather than a gap this paper first identifies. BurstGPT contributes production-derived workload characterization [16], and ServeGen studies

production workloads and their generation [17]. Workload resampling has a longer history in parallel-job scheduler evaluation [19]. Our analysis instead holds a selected trace corpus fixed, matches each adaptation on the same window and condition, and distinguishes source means from cross-window comparisons. We do not infer that the selected windows are independent draws from provider populations.

Serving simulators and model credibility. Vidur combines profiling and predictive models to explore LLM inference configurations [1]; LLMServingSim develops HW/SW co-simulation for serving workloads [7]. Those frameworks target hardware-aware performance estimation and evaluate prediction accuracy. CM-0.1 uses fixed-duration batches and an explicit work ledger to make adaptations comparable and execution independently checkable. It cannot replace their predictive objectives or borrow their validation evidence. Following the distinction between verification and validation in simulation methodology [14], reconstructing an execution journal tests whether the program implements the declared model. Establishing adequacy for a physical serving system would require separate evidence.

Statistical procedures. The calibration diagnostic uses existing procedures: Benjamini–Hochberg (BH), Benjamini–Yekutieli (BY), Holm, and bounded concentration inequalities [5, 6, 8, 9]. Their assumptions do not follow from successful code execution. In particular, a multiplicity adjustment cannot establish validity of its input p-values or repair a mismatch between the tested null and a practical performance claim. STAT01 investigates those issues in specified synthetic scenarios; it does not propose a replacement test.

3 A Common Execution Contract

3.1 Why the comparison object matters

The historical corpus contains 9,360 configurations and two deterministic verification passes per configuration. Its primary panel contains eleven policies, with two additional style approximations. The historical engine permits policy-dependent service modes: instantaneous prefill for some policies, explicit prefill work for others, and differing enforcement of KV limits. A policy's requested work can also differ from the engine's realized progress. Consequently, those outcomes cannot be interpreted as a controlled comparison of scheduling order. Table 1 distinguishes the three evidence objects used in this paper.

3.2 Service, resources, and information

CM-0.1 represents one abstract device. Each batch lasts 1 ms and has a 32,768-token work budget. Each executed prompt token consumes one unit; each selected decode request consumes one unit and emits one token. All policies share these rules. A prefill batch emits no output, including its final chunk: decoding can begin only in a later batch. Batch duration is independent of its composition. Thus seconds and tokens per second below are model quantities, not estimates for a particular GPU or LLM.

Each resident request reserves the complete 32,768-token context bound. All frozen request lengths fit; the maximum prompt-plus-output length is 30,053. Capacities of 131,072 and 262,144 tokens therefore permit *four and eight resident requests*, respectively, even though the configured sequence limit is 64. Actual used KV follows executed prompt and output prefixes, rounded to 16-token blocks, and must remain within the reservation. Completion releases the reservation. Waiting requests own none. This conservative rule avoids output-length prediction and allocation deadlock; it does not model incremental native paging or cache eviction.

Table 1. Evidence objects and the conclusions they support. Historical identifiers containing "faithful" are immutable names, not validation certificates.

Object	Comparison or check	Interpretation
Historical LSSP	Policy-dependent service/resource modes; archived ranking arithmetic	Motivation and historical-rule counts
C01-FULL	Four adaptations; common CM-0.1 execution and information	Descriptive finite-corpus performance
STAT01	Synthetic independent studies with known population truth	Scenario-specific inference calibration

Table 2. Project-maintained adaptations. Artifact IDs append `_c01` to `whole_fcfs`, `chunked_fcfs`, `decode_first`, and `slo_gated`. All use the shared contract above.

Adaptation	Historical lineage	Executed ordering rule and principal restriction
Whole-FCFS	vLLM-inspired whole-prompt implementation	Admit whole prompts in arrival order when feasible; otherwise decode residents. Admitting prefills defers all decodes that batch. No native paged allocator.
Chunked-FCFS	vLLM-inspired chunked implementation	Resident work before waiting work in stable order; at most 512 prefill tokens per request per batch.
Decode-first	Sarathi-inspired implementation	Resident decodes, then resident prefills, then waiting prefills; 512-token cap. No original engine timing.
SLO-gated	SLAI-inspired implementation	Urgent decodes first; then resident prefills, waiting prefills by prompt length, and residual decodes. Fixed 0.495 s gate; 512-token cap. No dynamic offset, heterogeneous classes, or full SLAI.

There is no preemption, dropping, migration, prefix sharing, or recomputation. Requests arriving during a batch become eligible at its end. Idle time advances to the next arrival. The administrative horizon is the final arrival plus 50 s; a batch cannot finish beyond it. Unfinished requests remain explicitly unfinished rather than contributing observed completion latencies.

Policies see only arrived request identifiers, stable order, arrival time, prompt length, acknowledged progress, residency, last-output/prefill-completion anchor, and current time. Actual remaining output length, future arrivals, source identity, and output-derived metric deadlines are hidden. The engine either executes the exact proposed token spans or rejects the plan. The shared contract therefore fixes both the resources and the online information available to each adaptation.

3.3 Adaptation identities and structural consequences

Table 2 defines the four evaluated adaptations. Published systems are lineage references; historical project implementations and CM-0.1 adaptations are separate objects. The panel was targeted using the historical comparison questions, not sampled to represent all schedulers.

Two consequences are essential to interpreting the results. At most eight requests can execute, and each chunked request uses at most 512 tokens. Their total work is therefore at most $8 \times 512 = 4,096$, below the shared 32,768-token budget. Every selected chunk and resident decode fits. Chunked-FCFS and decode-first consequently make the same per-request progress despite ordering their work differently. The complete saved terminal records confirm this equivalence in all 1,440 matched conditions.

The same bound means that SLO-gated's residual decodes fit every batch. After prefill completion, a resident's anchor advances with every output; its elapsed time cannot reach the 0.495 s urgency threshold in these executions. The distinctive behavior is principally prompt-length admission ordering with the frozen chunking rules. The results cannot be credited to an active urgency gate. Whole-FCFS can still defer decoding for admitted prefills; the work-bound equivalence does not apply to that rule.

3.4 Conformance evidence

The event referee reconstructs request states, token spans, timing, completion, and resource use from the inputs and execution journal without importing the engine or policy implementation. It checks that intended and executed work agree and recomputes the metrics. Every campaign cell passed this check; an independent analyzer also checked the complete ranking and reversal reports. The final evidence review additionally checked all result identities and terminal metrics and replayed 24 structurally selected new-cell journals. These checks establish conformance to CM-0.1, not GPU fidelity. Figure 1 summarizes the evidence chain.

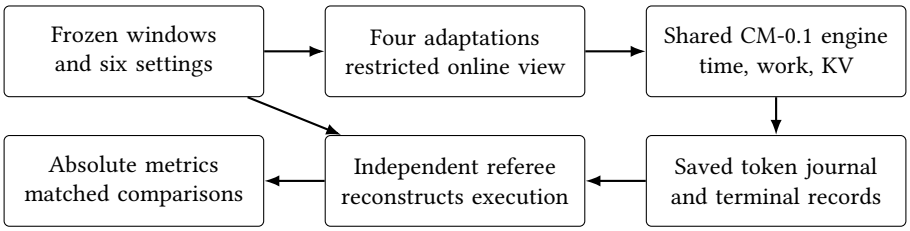


Fig. 1. Comparison and verification under one model. The independent analysis check recomputes aggregates, ranks, and reversals from the absolute metrics. No stage in this chain establishes physical-system accuracy.

4 Corpus, Conditions, and Analysis

4.1 Trace and implementation provenance

The corpus comprises 40 windows of 200 requests from each of Azure 2024, Bailian/Qwen, and BurstGPT [3, 12, 16]. These are three separately released sources, not three independent provider populations: BurstGPT and Azure have Azure infrastructure in their collection provenance. Table 3 identifies the consumed files and transformations. The supplement records full file and window fingerprints.

Each source's 40-window panel contains ten inherited pilot windows and 30 subsequent windows selected using fixed seeds and valid-row ranges. The expanded manifest was frozen before its corresponding campaign outcomes; earlier pilot outcomes already existed. Within the C01 manifest, row-overlap checks find no overlapping selected ranges. Exact overlap with some related projects' older BurstGPT selections remains unresolved. Nonoverlap within this panel does not establish temporal independence or representativeness. The frozen chronology labels allocate 20/10/10 windows per source to EARLY/MIDDLE/LATE. Four consecutive ten-window blocks provide an additional descriptive sensitivity; their elapsed time spans are unequal.

C01 reuses the historical input reconstruction, while its execution engine, policy views, and adaptation implementations have separate identities. C01-FULL retains C01 execution semantics at commit `9aed0206de66`; its full-campaign execution commit is `2d8342e699f3`. The complete matrix has $120 \times 6 \times 4 \times 2 = 5,760$ cells. Of these, 144 were reused from the pilot only after matching input, code, configuration, and journal identities; 5,616 were newly executed in the completed full

Table 3. Consumed dataset provenance. Each row contributes 40 windows and 8,000 request records. Hashes are prefixes of SHA-256 values; full values are in the bundled input manifest. License verification is a separate submission item, not implied by a hash.

Source	File and representation	File hash prefix
Azure 2024	`AzureLLMInferenceTrace_conv_2024.csv`; source arrival and token counts	`a0cc9b969a9b`
Bailian/Qwen	`qwen_traceB_blksz_16.jsonl`; input/output lengths read directly; 16-token prefix-block hashes are separate fields	`68e3f98e2d60`
BurstGPT	`BurstGPT_without_fails_2.csv`; retained valid source rows and token counts	`56193aa9b2bb`

campaign. Reuse is not an independent replication. No cell is omitted, and a deterministic cell is not rerun to manufacture a sampling distribution.

4.2 Inherited settings jointly transform arrivals and deadlines

The six labels LOW, PRE_KNEE, KNEE, POST_KNEE, OVERLOAD, and HIGH_PRESSURE come from historical calibration, not from a saturation analysis of CM-0.1. For a window-specific compression factor f, rebased arrival gaps and synthetic deadline slack are both divided by f. In idealized notation,

$$a'_i = (a_i - a_1)/f, \qquad d'_i = a'_i + (d_i - a_i)/f. \tag{1}$$

The implementation accumulates scaled nonnegative gaps in arrival order. Prompt/output lengths and priorities are unchanged. Factors range from 22.525 to 180.873 for Azure, 17.956 to 186.580 for Bailian, and 102.884 to 725.020 for BurstGPT. Equal labels therefore do not mean equal absolute arrival rates across windows or sources.

Deadlines are synthetic, derived in the inherited workload construction using output-length information; priorities are uniform. These attributes are not observed production service contracts. Deadlines are used only by the metric layer and are hidden from the scheduling adaptations. The setting axis is consequently *joint arrival/deadline stress*. A change in goodput across this axis cannot be attributed solely to arrival load.

4.3 Endpoints and aggregation

For arrival a_i, first output u_i, completion c_i, and output count o_i, latency is $c_i - a_i$ and time to first token (TTFT) is $u_i - a_i$. Time per output token (TPOT) is $(c_i - u_i)/(o_i - 1)$ for $o_i > 1$. The saved means and p95s use their defined completed-request populations; one-output requests do not have TPOT. The p95 uses the saved linear interpolation convention. Completion fraction is completed requests divided by all arrivals. Token throughput is completed output tokens divided by elapsed time from first arrival to final completion, or to the horizon if unfinished. Emitted-token throughput also counts any output of unfinished requests. Request throughput uses the same time denominator.

Arrival-normalized weighted goodput is

$$\mathrm{ANWG} = \frac{\sum_i w_i \, \mathbf{1}\{i \text{ completes and } c_i \le d'_i\}}{\sum_i w_i}. \tag{2}$$

Here $w_i = 1$, so ANWG is the fraction of arrivals meeting their synthetic deadline. It is dimensionless, unlike requests or tokens per second.

The eleven frozen ranking endpoints are ANWG, completion fraction, three throughput measures, and mean/p95 latency, TTFT, and TPOT. Larger is better for goodput, completion, and throughput; smaller is better for the six time endpoints. The complete endpoint set remains in the supplement, including redundant and numerically uninformative endpoints.

The window is the principal analysis unit. At a fixed source, setting, capacity, and policy, source means give each of the 40 windows equal weight. For the compact source–capacity table, we first average the six settings within each window and then average the 40 windows. A mean of window p95s is always labeled as such; it is not a pooled-request p95. Matched differences hold window, setting, and capacity fixed. We show all windows, not a selected favorable subset.

4.4 Ranking and practical-reversal definitions

Exact equality is primary: all maximizers or minimizers enter the top set, and tied policies receive average ranks. The frozen sensitivity admits values within $10^{-12}\max(1, |v_{\text{best}}|)$ of the best; it does not replace exact rankings or the practical-margin rule.

For policies A, B and condition c, let $\Delta_c = q_m(v_{A,c} - v_{B,c})$, where $q_m = 1$ for a larger-is-better endpoint and $q_m = -1$ otherwise. The relative margin is

$$r_c = \frac{|v_{A,c} - v_{B,c}|}{|v_{\text{loser},c}|}. \tag{3}$$

A practical reversal between c and c' requires $\Delta_c\Delta_{c'} < 0$ and both $r_c > 0.10$ and $r_{c'} > 0.10$. The strict boundary is evaluated exactly on the represented binary values. Zero losing values and missing values are unestimable. Nonzero ties remain eligible with margin zero. All denominators retain these distinctions.

Separate summaries compare within-source window pairs, Cartesian cross-source window pairs, settings within a window, capacities at a fixed window and setting, metrics at a fixed window, setting, and capacity, and source means at a fixed setting/capacity/metric. Cross-source windows are not paired by ordinal. For metric comparisons, each endpoint is oriented separately. These overlapping comparisons describe the finite corpus. We attach no population confidence intervals, p-values, or significance stars to C01.

5 Absolute Performance and Trade-offs (RQ1)

5.1 Completion and performance magnitudes

Every cell completes all 200 requests: 1,152,000 request executions across repeated conditions, not independent requests. There is no horizon censoring and no undefined cell-level latency, TTFT, or TPOT endpoint. TPOT populations range from 103 to 200 because single-output requests are excluded from that particular metric. Across cells, mean latency ranges from 0.008314 to 3.893923 s, p95 latency from 0.008560 to 7.300052 s, and token throughput from 332.489 to 7,766.685 tokens per simulated second.

Table 4 reports absolute source–capacity aggregates. Chunked-FCFS and decode-first share a row because their terminal records and all metrics are exactly equal; their separate identities remain in the artifact. Reporting this equivalence avoids treating duplicated outcomes as corroborating evidence from two different behaviors.

5.2 Mean latency gains do not imply a universal improvement

SLO-gated reduces aggregate mean latency against chunked-FCFS in all six source–capacity groups, by 11.3–34.0%. The largest aggregate reduction is Azure at four slots, from 2.8132 to 1.8559 s. At eight slots, BurstGPT changes from 0.1907 to 0.1692 s, the smallest percentage improvement. This

Table 4. Absolute performance, equally averaging six settings within each window and 40 windows within each source. "Slots" are effective full-context residents. Mean and p95 columns are mean latency and *mean window-level p95 latency*; throughput is output tokens per simulated second.

Source	Slots	Adaptation	ANWG	Mean (s)	p95 (s)	Tokens/s
Azure 2024	4	Chunked / Decode	0.1746	2.8132	5.3232	3746.2
		Whole-FCFS	0.1746	2.8145	5.3275	3743.5
		SLO-gated	0.2001	1.8559	5.1342	3684.4
Azure 2024	8	Chunked / Decode	0.3174	1.3626	2.5986	7018.7
		Whole-FCFS	0.3101	1.4092	2.6892	6816.5
		SLO-gated	0.3892	0.9357	2.5131	6825.8
Bailian/Qwen	4	Chunked / Decode	0.1828	2.2655	4.2969	3603.2
		Whole-FCFS	0.1799	2.3020	4.3705	3548.3
		SLO-gated	0.1771	1.7205	3.9681	3531.4
Bailian/Qwen	8	Chunked / Decode	0.3330	1.0417	2.0227	6183.4
		Whole-FCFS	0.3189	1.1063	2.1463	5914.1
		SLO-gated	0.3693	0.8087	1.9037	5935.7
BurstGPT	4	Chunked / Decode	0.1030	0.5225	1.0005	3354.9
		Whole-FCFS	0.0889	0.5654	1.0843	3054.9
		SLO-gated	0.2321	0.4448	1.0377	3223.4
BurstGPT	8	Chunked / Decode	0.3072	0.1907	0.3707	5532.1
		Whole-FCFS	0.2129	0.2314	0.4505	4796.5
		SLO-gated	0.4613	0.1692	0.4410	5322.3

is a substantial difference in the model's mean response time, but it comes with a throughput reduction in every group, ranging from 1.65% to 4.01%.

The tail comparison is less favorable. For BurstGPT, mean window-p95 latency rises from 1.0005 to 1.0377 s at four slots and from 0.3707 to 0.4410 s at eight slots: increases of 3.72% and 18.97%. Shorter average latency can therefore coexist with longer upper-tail latency under the same contract. At four-slot Bailian, ANWG is also 3.12% lower for SLO-gated despite its 24.05% mean-latency reduction. Throughput, deadline attainment, and response time reward different aspects of the completed schedule.

Because the urgency gate does not bind (Section 3), these observations concern prompt-length admission ordering and the declared chunking rules. Earlier admission of short prompts can change who waits and which requests finish last. The measured mean/tail/throughput contrast is consistent with that ordering distinction, but the study does not isolate separate causal contributions by intervention or validate the original SLAI system.

5.3 Capacity and inherited-setting effects

Increasing reservation capacity doubles permitted residency from four to eight. In every source–adaptation aggregate shown, mean and mean-p95 latency fall and throughput rises. For chunked-FCFS, Azure mean latency falls from 2.8132 to 1.3626 s while throughput rises from 3,746.2 to 7,018.7 tokens/s; BurstGPT changes from 0.5225 to 0.1907 s and from 3,354.9 to 5,532.1 tokens/s. The capacity axis is predominantly a concurrency constraint in this model. The largest observed used KV is 47,872 tokens in the four-slot profile and 62,880 in the eight-slot profile, both below even the smaller nominal capacity. These results do not measure native allocator pressure, eviction, or the benefit of paging at an exhausted used-token limit.

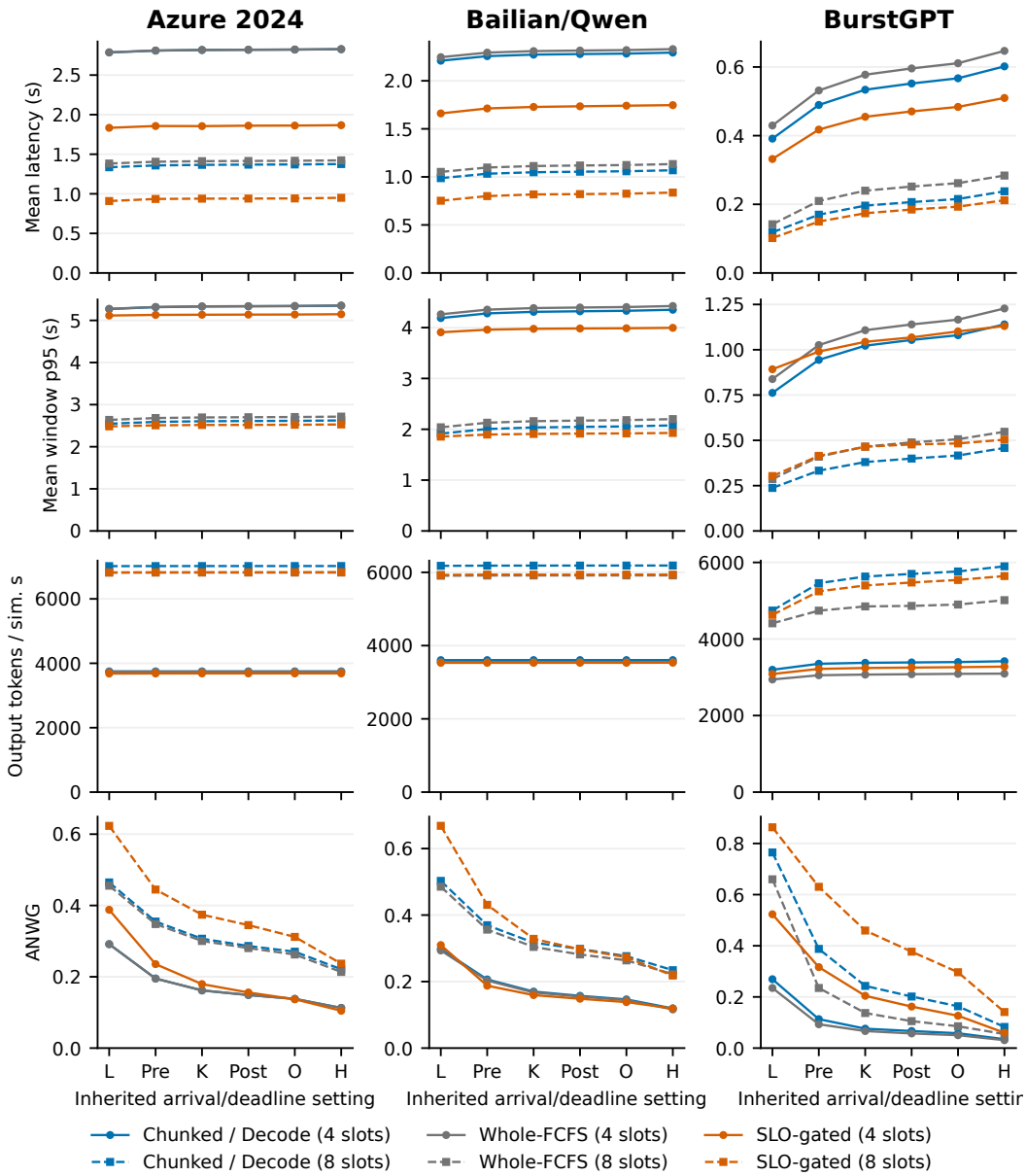


Fig. 2. Absolute performance across all six inherited settings, equally averaged over 40 windows per source. Solid circles: four slots; dashed squares: eight slots. L/Pre/K/Post/O/H abbreviate LOW through HIGH_PRESSURE in their declared order. Chunked and decode-first curves coincide exactly. The p95 panels average window-level p95s. Both arrivals and synthetic deadlines change along the horizontal axis. Lines connect descriptive condition means, with no inferential uncertainty bands.

Figure 2 separates settings within each source–capacity. The inherited setting labels can produce large changes in goodput with small changes in latency or throughput. For Azure chunked-FCFS at four slots, LOW to HIGH_PRESSURE changes mean latency from 2.7858 to 2.8269 s and throughput from 3,745.9 to 3,746.3 tokens/s, while ANWG falls from 0.291375 to 0.112625. Since the transformation also contracts deadline slack, this is not evidence for an isolated arrival-load effect on goodput.

5.4 Matched-window variation

Figure 3 displays all 40 window-level matched differences per source and capacity, averaging the six paired setting differences within each window. It complements the source means by showing where a performance difference changes sign or is concentrated. The vertical zero line represents equal performance; it is neither a confidence bound nor the 10% practical-reversal threshold. For latency, a negative difference favors SLO-gated; for throughput, it favors chunked-FCFS.

At the unaveraged condition level, SLO-gated uniquely minimizes mean latency in all 480 Azure conditions, 468 of 480 Bailian conditions, and only 198 of 480 BurstGPT conditions. Its favorable source aggregates therefore do not mean that it wins every window/setting comparison. The tail and throughput distributions make the same distinction visible without reducing each comparison to a winner label.

Answer to RQ1. Absolute performance varies with workload and permitted residency, and SLO-gated's aggregate mean-latency advantage coexists with throughput losses and BurstGPT tail-latency losses. Completion is uniformly one. The capacity contrast primarily reflects four versus eight full-context residents, and the setting contrast jointly changes arrival and deadline stress. Two nominal adaptations have identical behavior under the evaluated contract.

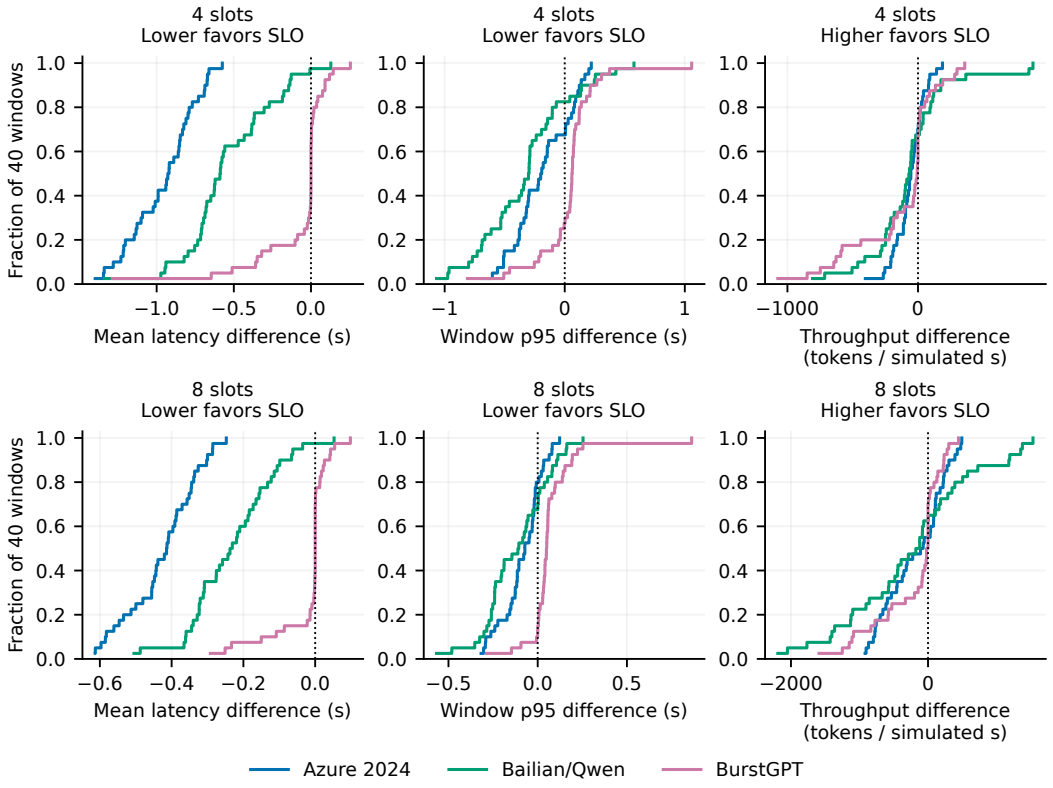


Fig. 3. Empirical distributions of SLO-gated minus chunked-FCFS differences. Each curve has 40 equally weighted windows; each point averages six matched setting differences within one window. The axes show absolute differences, not percentages. Positive latency differences and negative throughput differences are unfavorable to SLO-gated. All windows are retained.

6 Rankings, Aggregation, and Metric Choice (RQ2)

6.1 Stable practical source comparisons and heterogeneous windows

Table 5 reports the strict-10% reversal rule with eligible and total denominators. Across all source-mean comparisons the count is **zero out of 2,376**. Each source mean uses all 40 windows at a fixed setting, capacity, and metric. The denominator is 3 source pairs ×6 settings ×2 capacities ×11 endpoints ×6 policy pairs. The common-model study thus does not reproduce a broad claim of practical source-mean ranking reversals.

This null result is compatible with smaller sign changes and different tie sets. Among 396 comparisons of source pairs at fixed settings, capacities, and metrics, exact top sets are equal in 307 and intersect in 323; complete average-rank vectors are equal in 247. Identical practical ordering and identical full rankings are different statements.

Window-level variation remains visible. Within-source window comparisons yield 34,394 reversals among 1,847,700 eligible comparisons; Cartesian cross-source window comparisons yield 109,913 among 3,788,640. For ANWG alone, the corresponding counts are 8,472/162,900 and 35,352/332,640. The larger window-pair denominators arise from comparing every unordered pair of 40 windows within a source (780) or every Cartesian pair across sources (1,600). They are

Table 5. Strict-10% practical reversals. Nonzero ties remain in eligible denominators; a zero losing value is unestimable. All counts include six policy pairs and the declared endpoints. Reused windows and overlapping pairs make these dependent descriptive counts, not independent trials.

Comparison axis	Reversals	Eligible	Total	Unestimable
Within-source windows	34,394	1,847,700	1,853,280	5,580
Cross-source windows	109,913	3,788,640	3,801,600	12,960
Inherited settings	920	236,853	237,600	747
KV capacities	350	47,418	47,520	102
Metric pairs	11,795	473,580	475,200	1,620
Source means	0	2,376	2,376	0

Table 6. Ranking summaries, each out of 1,440 window/setting/capacity conditions. "Near-tie change" counts conditions whose top set changes under the predeclared $10^{-12}\max(1, |v_{\text{best}}|)$ tolerance. No ranking is missing. "SLO unique top" always uses exact equality.

Endpoint	Exact top tie	SLO unique top	Near-tie change
ANWG	425	1,004	0
Mean latency	289	1,146	6
p95 latency	586	834	0
Token throughput	906	482	0
Completion	1,440	0	0
Mean TPOT	979	461	1,129
p95 TPOT	1,348	92	104

not larger independent sample sizes. These summaries cannot establish that one axis causally contributes more instability than another.

Within the same window, settings produce 920/236,853 eligible reversals, capacity changes 350/47,418, and metric choice 11,795/473,580. The last count compares policy-pair directions across endpoints; it demonstrates that a preference can depend on the objective even with the workload and resources fixed. It is distinct from comparing top sets, which asks whether at least one best policy is shared rather than whether a particular pair reverses.

6.2 Ties and redundant endpoints

Table 6 retains exact ties instead of breaking them by policy identifier. A unique mean-latency winner is not necessarily a unique throughput winner: SLO-gated is uniquely best in 1,146 and 482 conditions, respectively. Completion has a four-way tie everywhere. Treating an arbitrary deterministic tie-break as a scientific winner would discard that information.

Several endpoints provide the same ranking mechanically. All requests finish, so emitted-token and completed-token throughput are identical. At a fixed condition, request throughput differs from token throughput only by a workload-specific multiplier shared by every policy. All three throughput rankings therefore coincide. Counting them as independent evidence would exaggerate the breadth of a finding.

TPOT supplies a different warning. Chunked-FCFS, decode-first, and SLO-gated produce essentially one output per millisecond once decoding begins. The largest chunked-versus-SLO difference in cell mean TPOT is 1.80×10^{-16} s. Floating-point timestamp subtraction can still change an exact winner. The near-tie sensitivity changes mean-TPOT top sets in 1,129/1,440 conditions and

p95-TPOT top sets in 104/1,440. These differences carry no practical performance significance. The original exact values remain preserved, alongside the predeclared sensitivity.

Across all 79,200 metric-pair ranking comparisons, exact top sets are equal in 27,915 and intersect in 57,602. These counts include redundant throughput endpoints, completion ties, and numerical TPOT distinctions. They summarize reporting consequences; they are not eleven independent tests of a scheduler's robustness.

6.3 Chronological summaries

Figure 4 reports SLO-minus-chunked ANWG differences in four consecutive ten-window blocks. Across settings and capacities, block means range from −0.0120 to +0.2030 for Azure, −0.0300 to +0.2330 for Bailian, and +0.0150 to +0.3275 for BurstGPT. Negative differences occur in 8/48 Azure and 24/48 Bailian block/setting/capacity combinations, and none of the 48 BurstGPT combinations. The frozen EARLY/MIDDLE/LATE summaries show the same qualitative contrast: negative differences in 6/36, 20/36, and 0/36 combinations, respectively. These are corpus-order sensitivities, not estimates of a temporal process or guarantees for future traffic.

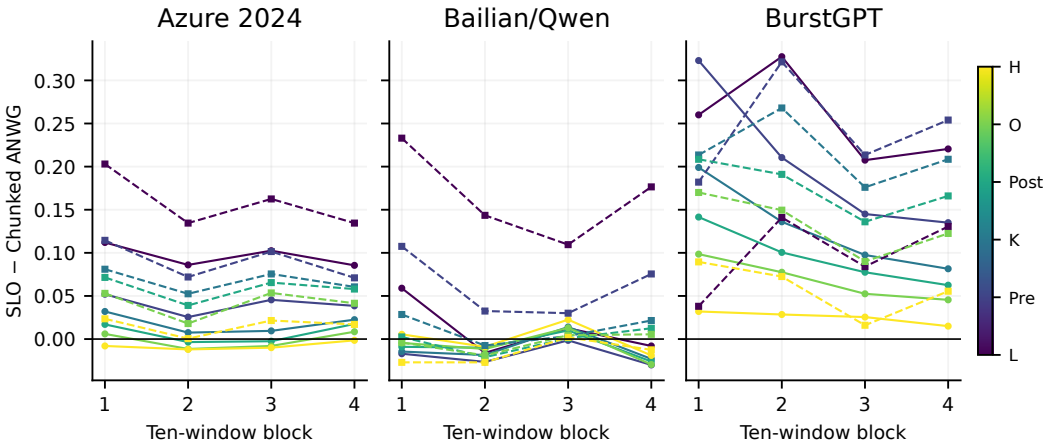


Fig. 4. Chronological block means of SLO-gated minus chunked-FCFS ANWG. Each block contains ten frozen windows. Colors identify inherited settings (L through H); solid circles and dashed squares denote four and eight slots. Blocks have unequal durations and are not independent temporal samples. The figure uses all 144 block/setting/capacity combinations.

Answer to RQ2. No source-mean policy comparison reverses by more than 10% in both directions. Window-level and metric-dependent reversals nevertheless occur. Exact ties, duplicated throughput rankings, and negligible TPOT differences show why absolute magnitudes and top sets are needed alongside ranks. Aggregation changes the question being answered; stability at the source-mean level does not imply uniform preferences across its constituent windows.

7 Historical Interpretation and Calibration (RQ3)

7.1 What the historical comparison establishes

The historical analysis reported 36 of 990 policy-pair/source-pair/setting records meeting its practical-margin and nominal-interval rule. Numerical reproduction confirms that archived-rule count. It does not establish population reversals: the historical policy–model combinations differ,

Table 7. Compact calibration diagnostic for the actual historical-source screened-BH comparator. Each row has 8,000 studies; ordinary fixed-contrast interval coverage is available in all of them. Bounds are the expanded 17-method simultaneous Monte Carlo bounds, not confidence intervals for C01 performance. These three null scenarios isolate IID, independent-block, and AR dependence; all 18 scenarios and methods remain in the supplement.

Scenario	FDR	MC bounds	Coverage	MC bounds
IID equal null	0.09025	[0.06878, 0.11172]	0.94187	[0.92357, 0.96018]
Independent-block null	0.91800	[0.89726, 0.93874]	0.64762	[0.61465, 0.68060]
AR(0.7) null	0.98037	[0.96780, 0.99295]	0.56487	[0.53081, 0.59894]

selected trace windows have no known population sampling law, and the inference procedure's calibration was unestablished. The C01 panel and execution model change the comparison object, so its zero source-mean practical reversals are not a like-for-like disappearance of those 36 records. Historical and common-model outcomes are kept in separate evidence strata rather than combined into a leaderboard.

7.2 A focused known-truth diagnostic

STAT01 evaluates historical inference in synthetic studies whose population means, covariance, and missingness are specified. Each study has three sources, eleven policies, and normally 40 windows per source; one bounded scenario uses 400. The 165 source-pair/policy-pair hypotheses are dependent within a study. Independently generated studies, not their individual hypotheses, are the Monte Carlo units. The completed design contains 134,000 studies across 18 scenarios, including IID, block-dependent, autoregressive, nonnormal, bounded, and missing-data cases.

The primary truth is a population reversal exceeding 10% relative to the losing mean in both sources. Sign reversals are a separate endpoint. The historical procedure first screens noisy point estimates at 10%, then uses bootstrap intervals and zero-effect p-values on the selected contrasts. Testing a zero-effect null after such screening is not automatically a test of the practical-effect null. BH or BY alone cannot supply the missing calibration [5, 6].

Three implementation identities are retained. The eight frozen original procedures include a port of the historical screened-BH procedure. A post-diagnosis comparator reproduces the actual historical source's floating-point evaluation order. Eight additional comparisons implement explicit mathematical threshold semantics. Thus the expanded family has 17 methods, with nine additions identified as post-diagnosis. Arithmetic correction is not a new test and does not change the inferential assumptions. Appendix A gives the definitions and the saved boundary case.

Table 7 shows that the actual historical-source procedure has practical FDR 0.09025 in the IID equal-null scenario, with an expanded Monte Carlo lower bound above 0.05. Independent-block and AR nulls show much larger false-discovery rates and poor ordinary interval coverage. These null examples already demonstrate a calibration concern without requiring interpretation of power or near-boundary alternatives. Other scenarios expose the practical-versus-sign target mismatch: practical FDR is 0.39500 at the exact practical boundary and approximately 0.73316 in the medium-reversal scenario. Full scenario results are retained rather than summarized as a single global error rate.

These findings also have limits. Assumption-aligned Gaussian procedures frequently remain *inconclusive* under the stringent simultaneous Monte Carlo screen; that is not evidence that they are miscalibrated. Conservative bounded Hoeffding–Holm comparisons have zero observed false discoveries and 100% observed diagnostic coverage, but practical power is zero at 40 windows

and approximately 0.05263 at 400. The campaign does not certify a generally useful replacement procedure. Of 144 original method/scenario classifications, 45 indicate concern, 74 are inconclusive, 9 meet the empirical tolerance screen, and 16 abstain outside their domain. Seven of the nine compatible entries arise in the degenerate exact-ties scenario; the remaining two are bounded Hoeffding.

Completion and precision must also be separated. Scenarios 3 and 17 stop at 2,000 and 4,000 studies; the other sixteen finish at 8,000. Six scenarios meet the original precision objective, while twelve stop at the maximum count with precision gaps. The original eight-method uncertainty family controls stopping. The expanded 17-method report has its own, wider allocation; additions do not retrospectively become prespecified. Neither maximum-count completion nor an inconclusive calibration screen is a successful calibration claim.

Answer to RQ3. Reproducing historical arithmetic establishes what its rule reported; execution conformance establishes that C01 follows its common model. Neither establishes valid statistical inference. STAT01 demonstrates calibration failures for historical procedures in explicit known-truth scenarios, while retaining uncertainty and assumption-specific limits. Those synthetic results do not automatically validate or invalidate every trace-specific historical conclusion. C01's finite-corpus descriptions do not depend on population significance tests.

8 Discussion and Threats to Validity

What an evaluator should report. The useful comparison object is an adaptation together with its execution, resource, and information contract. A common contract exposes both nontrivial trade-offs and distinctions that cannot affect execution. Here, the budget and reservation rules collapse chunked-FCFS and decode-first to one behavior and leave the urgency gate inactive. Reporting four implementation names without these consequences would overstate mechanistic coverage. Absolute latency and throughput then reveal costs that a goodput-only or mean-latency-only ranking would miss.

Aggregation must be specified with the same care. A source mean answers a question about equal-window average performance; a cross-window reversal asks whether two particular windows prefer different policies. Neither supersedes the other. A reader selecting a scheduler for variable traffic needs both the aggregate magnitude and its window-level variation. Exact top sets retain ties, and a practical margin prevents negligible numerical differences from becoming performance claims.

Model adequacy and external validity. CM-0.1 deliberately fixes batch duration, assigns equal work units to prompt and decode tokens, reserves a full context before execution, and omits preemption and migration. It does not model kernel costs, bandwidth, communication, native allocation, model-dependent token timing, or changing batch efficiency. Its latency and throughput values describe the abstraction. Independent event conformance and deterministic replay cannot establish physical prediction accuracy [14]. Related native probes and other projects' mechanism studies do not validate this four-adaptation ranking matrix. No deployment recommendation follows directly from it.

Corpus and construct validity. The 120 selected windows are finite and temporally related. They represent particular files, valid-row filtering, and sampling rules rather than random samples of providers. The two Azure-linked sources should not be treated as independent providers, and cross-project overlap of some older BurstGPT selections is unresolved. Uniform priorities and synthetic, output-derived deadlines restrict the interpretation of ANWG. The inherited setting transformation changes deadlines together with arrivals. Full-context reservation capacity measures

a residency constraint, and the panel excludes many mechanisms available to production engines. These restrictions bound which workload and resource questions the study answers.

Verification and inferential validity. Independent checks reduce the risk that reported metrics disagree with the model or saved records; they cannot remove assumptions shared by the model and the scientific question. The final review checked every result identity and metric summary, while fresh journal replay sampled 24 new cells in addition to the exhaustive in-campaign referee. C01's descriptive scope avoids unsupported sampling assumptions but does not imply population robustness. STAT01 supplies known truth only for its specified generators. Its post-diagnosis additions, finite study counts, conditional interval availability, and conservative power must remain visible when interpreting calibration screens.

9 Conclusion

Under a common execution contract, scheduler comparison is as much about performance magnitudes and inactive distinctions as about the rank order. For RQ1, SLO-gated reduces aggregate mean latency by 11.3–34.0% but loses throughput, with worse BurstGPT mean window-p95 latency; chunked-FCFS and decode-first are identical throughout the evaluated matrix. For RQ2, no source-mean comparison has a strict 10% practical reversal, while window and metric comparisons do, and many exact rankings are tied or redundant. For RQ3, the historical audit and synthetic diagnostic distinguish reproducible execution and arithmetic from calibrated inference. The result is a bounded, traceable performance evaluation: its strongest conclusions concern the measured trade-offs and invariances of these adaptations under CM-0.1.

Artifact Availability and Author-Review Items

The M3 review package includes source, figures, tables, saved analysis summaries with SHA-256 fingerprints, generation scripts, and both ACM manuscript and `acmsmall` PDFs. Supplementary CSV files retain all 5,760 C01 cells, chronological aggregates, reversal groups, matched-window differences, and all 7,200 final STAT01 endpoint summaries across the original and expanded families. Saved inputs are copied without modification; M2 and experimental artifacts remain preserved. The packet supports regeneration of the paper's displays from saved results. Full execution reproduction additionally requires the pinned code, request inputs, and journals identified in the evidence ledger.
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AUTHOR REVIEW REQUIRED: Codex assisted the M3 rewrite, saved-data display generation, and validation reports. Confirm the final AI-assistance disclosure and review responsibility under current publisher policy. Confirm current official ToMPECS submission requirements, including

format, anonymity, length, and supplementary-material handling. The official author-guidelines page was inaccessible during preparation; no ToMPECS-specific page limit is assumed.

References

[1] Amey Agrawal, Nitin Kedia, Jayashree Mohan, Ashish Panwar, Nipun Kwatra, Bhargav S. Gulavani, Ramachandran Ramjee, and Alexey Tumanov. 2024. VIDUR: A Large-Scale Simulation Framework for LLM Inference. In *Proceedings of Machine Learning and Systems (MLSys '24)*, Vol. 6. MLSys, 351–366. https://proceedings.mlsys.org/paper_files/paper/2024/hash/b74a8de47d2b3c928360e0a011f48351-Abstract-Conference.html

[2] Amey Agrawal, Nitin Kedia, Ashish Panwar, Jayashree Mohan, Nipun Kwatra, Bhargav S. Gulavani, Alexey Tumanov, and Ramachandran Ramjee. 2024. Taming Throughput-Latency Tradeoff in LLM Inference with Sarathi-Serve. In *Proceedings of the 18th USENIX Symposium on Operating Systems Design and Implementation (OSDI '24)*. USENIX Association, Santa Clara, CA, 117–134. https://www.usenix.org/conference/osdi24/presentation/agrawal

[3] Alibaba Cloud. 2026. *qwen-bailian-usagetraces-anon.* https://github.com/alibaba-edu/qwen-bailian-usagetraces-anon/tree/5f7439c51ec248a0c585f7d90a41a6f57773b912 Anonymized Bailian/Qwen production LLM-serving usage-trace release; repository `alibaba-edu/qwen-bailian-usagetraces-anon`, pinned commit `5f7439c51ec248a0c585f7d90a41a6f57773b912`; release-specific licensing requires author verification.

[4] Agrim Bari, Parikshit Hegde, and Gustavo de Veciana. 2025. Optimal Scheduling Algorithms for LLM Inference: Theory and Practice. *Proceedings of the ACM on Measurement and Analysis of Computing Systems* 9, 3, Article 59 (2025), 43 pages. doi:10.1145/3771574

[5] Yoav Benjamini and Yosef Hochberg. 1995. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society: Series B (Methodological)* 57, 1 (1995), 289–300. doi:10.1111/j.2517-6161.1995.tb02031.x

[6] Yoav Benjamini and Daniel Yekutieli. 2001. The Control of the False Discovery Rate in Multiple Testing under Dependency. *The Annals of Statistics* 29, 4 (2001), 1165–1188. doi:10.1214/aos/1013699998

[7] Jaehong Cho, Minsu Kim, Hyunmin Choi, Guseul Heo, and Jongse Park. 2024. LLMServingSim: A HW/SW Co-Simulation Infrastructure for LLM Inference Serving at Scale. arXiv:2408.05499v1. arXiv:2408.05499 https://arxiv.org/abs/2408.05499v1

[8] Wassily Hoeffding. 1963. Probability Inequalities for Sums of Bounded Random Variables. *J. Amer. Statist. Assoc.* 58, 301 (1963), 13–30. doi:10.1080/01621459.1963.10500830

[9] Sture Holm. 1979. A Simple Sequentially Rejective Multiple Test Procedure. *Scandinavian Journal of Statistics* 6, 2 (1979), 65–70. https://www.jstor.org/stable/4615733

[10] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. 2023. Efficient Memory Management for Large Language Model Serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP '23)*. ACM, Koblenz, Germany, 611–626. doi:10.1145/3600006.3613165

[11] Andreas Maurer and Massimiliano Pontil. 2009. Empirical Bernstein Bounds and Sample Variance Penalization. In *Proceedings of the 22nd Annual Conference on Learning Theory*. 9 pages. https://www.cs.mcgill.ca/~colt2009/papers/012.pdf

[12] Microsoft Azure. 2024. *AzureLLMInferenceDataset2024.* https://github.com/Azure/AzurePublicDataset/blob/master/AzureLLMInferenceDataset2024.md

[13] Konstantinos Papaioannou and Thaleia Dimitra Doudali. 2024. The Importance of Workload Choice in Evaluating LLM Inference Systems. In *Proceedings of the 4th Workshop on Machine Learning and Systems (EuroMLSys '24)*. ACM, Athens, Greece, 39–46. doi:10.1145/3642970.3655823

[14] Robert G. Sargent. 2013. Verification and validation of simulation models. *Journal of Simulation* 7, 1 (2013), 12–24. doi:10.1057/jos.2012.20

[15] Biao Sun, Ziming Huang, Hanyu Zhao, Wencong Xiao, Xinyi Zhang, Yong Li, and Wei Lin. 2024. Llumnix: Dynamic Scheduling for Large Language Model Serving. In *Proceedings of the 18th USENIX Symposium on Operating Systems Design and Implementation (OSDI '24)*. USENIX Association, Santa Clara, CA, 173–191. https://www.usenix.org/conference/osdi24/presentation/sun-biao

[16] Yuxin Wang, Yuhan Chen, Zeyu Li, Xueze Kang, Yuchu Fang, Yeju Zhou, Yang Zheng, Zhenheng Tang, Xin He, Rui Guo, Xin Wang, Qiang Wang, Amelie Chi Zhou, and Xiaowen Chu. 2025. BurstGPT: A Real-World Workload Dataset to Optimize LLM Serving Systems. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '25), Volume 2*. ACM, Toronto, ON, Canada, 5831–5841. doi:10.1145/3711896.3737413

[17] Yuxing Xiang, Xue Li, Kun Qian, Yan Zhang, Wenyuan Yu, Ennan Zhai, Xin Jin, and Jingren Zhou. 2026. ServeGen: Workload Characterization and Generation of Large Language Model Serving in Production. In *Proceedings of the 23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI '26)*. USENIX Association, Renton, WA,

1845–1859. https://www.usenix.org/conference/nsdi26/presentation/xiang-servegen

[18] Gyeong-In Yu, Joo Seong Jeong, Geon-Woo Kim, Soojeong Kim, and Byung-Gon Chun. 2022. Orca: A Distributed Serving System for Transformer-Based Generative Models. In *Proceedings of the 16th USENIX Symposium on Operating Systems Design and Implementation (OSDI '22)*. USENIX Association, Carlsbad, CA, 521–538. https://www.usenix.org/conference/osdi22/presentation/yu

[19] Netanel Zakay and Dror G. Feitelson. 2014. Workload resampling for performance evaluation of parallel job schedulers. *Concurrency and Computation: Practice and Experience* 26, 12 (2014), 2079–2105. doi:10.1002/cpe.3240

[20] Yinmin Zhong, Shengyu Liu, Junda Chen, Jianbo Hu, Yibo Zhu, Xuanzhe Liu, Xin Jin, and Hao Zhang. 2024. DistServe: Disaggregating Prefill and Decoding for Goodput-Optimized Large Language Model Serving. In *Proceedings of the 18th USENIX Symposium on Operating Systems Design and Implementation (OSDI '24)*. USENIX Association, Santa Clara, CA, 193–210. https://www.usenix.org/conference/osdi24/presentation/zhong-yinmin

A Statistical Diagnostic: Definitions and Complete Scenario Summary

A.1 Study construction and truth

A study draws a latent array $Y \in \mathbb{R}^{3\times n\times 11}$, with positive population means μ_{sj}. Sources are independent within this synthetic design. Dependence across policies is $C(\rho) = (1-\rho)I + \rho\mathbf{1}\mathbf{1}^{\mathsf{T}}$, usually with $\rho = 0.5$. Temporal covariance is identity, independent five-window blocks with within-block correlation 0.7, or AR(1) with correlation $0.7^{|i-j|}$. Source-specific innovations have mean zero and variance one; a Cholesky transformation imposes the specified covariance, with default scale $\sigma = 0.15$. The exact-ties scenario uses $\rho = 1$.

Each family comprises $\binom{3}{2}\binom{11}{2} = 165$ hypotheses. For a source pair (s,t) and policy pair (a,b), sign-reversal truth is $(\mu_{sa} - \mu_{sb})(\mu_{ta} - \mu_{tb}) < 0$. Practical-reversal truth is

$$\begin{aligned}(\mu_{sa} > 1.1\mu_{sb} \ \wedge \ \mu_{tb} > 1.1\mu_{ta}) \\ \vee \ (\mu_{sb} > 1.1\mu_{sa} \ \wedge \ \mu_{ta} > 1.1\mu_{tb}).\end{aligned} \tag{4}$$

Equality at 10% is null. Truth is enumerated from decimal/rational mean specifications, independently of estimated means and test decisions.

Equal-null means are 0.5. Stable and near-tie means are $0.5 + 0.01j$ and $0.5 + 0.0002j$, respectively. A reversal of size δ assigns source 0 policies 0/1 means $0.5 \pm \delta/2$, swaps them in source 1, and sets all other means to 0.5. Small, medium, and large IID reversal scenarios use $\delta = 0.03, 0.10, 0.30$. The boundary scenario uses 0.55/0.50 and the reverse. Block, AR, unequal-variance, mixed-distribution, and MCAR reversal scenarios use $\delta = 0.15$. Unequal scales are $(0.08, 0.25, 0.15)$; mixed innovations are standardized normal, Student-t_5, and lognormal. MCAR retains each policy/window observation independently with probability 0.8. Selective missingness removes source 0/policy 0 and source 1/policy 1 observations below their respective latent means; truth remains the unconditional mean. Observed values cannot identify that target merely by ignoring missingness.

The bounded scenarios use IID standardized uniform innovations, $\sigma = 0.08$, and $\delta = 0.30$, giving scores in $[0,1]$ analytically, without clipping. All scenarios have 40 windows per source except the last, which has 400. Raw arrays and masks are retained in the full artifact. The 112,275 reused and 21,725 subsequently generated studies sum to the completed 134,000-study design; no study was removed because of a numerical disagreement between implementations.

A.2 Frozen methods and post-diagnosis additions

All methods use nominal level $\alpha = q = 0.05$. The historical screened procedure uses 2,000 bootstrap draws of matched observed per-window policy differences, percentile intervals, and $p = \min\{1, 2\min[\Pr_{\text{boot}}(\bar{D} \le 0), \Pr_{\text{boot}}(\bar{D} \ge 0)]\}$. It combines source-side p-values by their maximum, applies BH to the point-screened family, and requires both intervals to exclude zero. The full-family variant assigns untested hypotheses p-value one and applies BH over all 165. The historical plus-one BY variant transforms the component bootstrap p-value as $\min\{1, (2000p + 2)/2001\}$ before source combination and applies BY over 165. These modifications do not make selection or the original p-values valid by definition.

The analytic practical-effect methods use paired contrasts $Z = Y_a - 1.1Y_b$ and their reversed orientation. Each orientation combines the two one-sided source tests by a maximum (an intersection–union test). The union over orientations uses $\min\{1, 2\min(p_1, p_2)\}$ before Holm correction over 165 hypotheses [9]. IID t-Holm uses paired Student tests; cluster t-Holm uses independent five-window cluster averages, retaining only completely observed clusters. Fewer than two complete clusters causes abstention. Their finite-sample Gaussian justifications apply to their respective independent sampling units with complete or MCAR observations, not arbitrary temporal dependence. The covariance-oracle normal variant uses the known Gaussian covariance of each observed

matched sample mean. It does not access missing latent outcomes and is not deployable without that covariance information.

Grouped-bootstrap BY resamples nonoverlapping five-window groups jointly across policies, preserving masks and pairing. Its centered one-sided component p-value is $(1+\#\{\bar{Z}^* - \bar{Z} \geq \bar{Z}\})/2001$. A zero-observation resample causes component abstention; it is not silently excluded. The same orientation construction is followed by full-family BY. No finite-sample guarantee was prespecified for this procedure. Bounded Hoeffding–Holm applies only to the bounded IID scenarios. Since $Z \in [-1.1, 1]$, its component p-value is $\exp[-2m \max(\bar{Z}, 0)^2/2.1^2]$ [8], followed by the same orientation and Holm construction. It abstains outside that domain.

The eight original identities are retained unchanged. The ninth method, `legacy_source_screened_BH`, reproduces the actual frozen historical source's arithmetic. Eight `exact_` counterparts of the original methods use explicit rational critical thresholds and exact bootstrap-count grids. Analytic p-values are treated as the exact dyadic values represented by their finite floats, not as infinitely accurate analytic probabilities. Rational harmonic constants define BY thresholds. These additions change numerical semantics, not the sampling assumptions, screening target, or statistical validity of a component test.

A.3 Coverage, availability, power, and Monte Carlo accounting

For each independent study, let R be discoveries and V false discoveries. False-discovery proportion is $V/\max(R, 1)$; its expectation is FDR. Familywise error is $\mathbf{1}\{V > 0\}$, and power is true discoveries relative to the number of alternatives when alternatives exist. Both practical and sign truth are retained. Ordinary interval coverage uses a fixed source 0/policy 0-minus-policy 1 contrast. For historical methods, that interval is generated in a separate unconditional diagnostic stream. Coverage of actually selected intervals is a different, selection-conditional endpoint; it is not replaced by ordinary coverage.

Interval availability is reported separately. In the MCAR and selectively missing scenarios, cluster-t ordinary coverage is based on 1,676 and 202 of 8,000 studies, respectively. Historical selected-interval coverage in the IID equal null is 0.123834 over 958 studies emitting selected intervals, whereas ordinary fixed-contrast coverage is 0.941875 over 8,000 studies. These are study-level averages, not independent interval observations. The complete endpoint CSV also includes null rejection fractions, maximizer-set coverage and size, and deterministic-ID correctness; exact empirical maximizing sets are not confidence sets.

The five possible analysis looks are 500, 1,000, 2,000, 4,000, and 8,000 studies per scenario. For a bounded study-level endpoint with sample variance s^2 and available count N, the two-sided empirical Bernstein half-width is

$$b = \sqrt{\frac{2s^2 \log(4/\delta)}{N}} + \frac{7 \log(4/\delta)}{3(N-1)}. \tag{5}$$

The bounds clip $\bar{X} \pm b$ to $[0, 1]$; empty endpoints are unestimated with the full range. The construction uses the empirical Bernstein bound and a finite union over both tails and all reported looks [11]. Conditional coverage uses the count of IID studies emitting an interval, with that availability explicit. The original allocation is $\delta = 0.05/(18 \cdot 8 \cdot 16 \cdot 5)$; the expanded report uses $0.05/(18 \cdot 17 \cdot 16 \cdot 5)$. Dependence among methods and endpoints does not invalidate the union bound. The wider accounting does not erase post-diagnosis method selection.

Stopping uses only the original eight methods. Required half-widths are 0.025 for practical FDR, familywise error, and null rejection fraction; 0.03 for ordinary interval coverage; and 0.08 for defined power. Selected-interval coverage is not a stopping target. The calibration screen flags concern if the practical-FDR lower bound exceeds 0.05 or ordinary-coverage upper bound is below

0.95. Empirical compatibility with tolerance requires FDR upper bound at most 0.06 and coverage lower bound at least 0.94 in relevant assumption-valid scenarios. All other estimable cases are inconclusive. The screen permits conservative overcoverage and is not a proof of exact nominal calibration.

Table 8. Complete original-eight scenario screen and stopping counts. C: calibration concern; I: inconclusive; E: empirical compatibility with tolerance; A: outside-domain abstention. P: original precision objective met; M: maximum-count stop with gaps. H-S/H-F/H-BY: historical screened BH/full BH/plus-one BY; t/c-t/O/G/Hf: IID t/cluster t/oracle normal/grouped bootstrap/bounded Hoeffding. These use the *original-eight* uncertainty family. Full numerical results for all 17 methods and both families are in `stat01-all-endpoints.csv`.

ID	Scenario	Studies	Stop	H-S	H-F	H-BY	t	c-t	O	G	Hf
0	iid_equal_null	8,000	P	C	I	I	I	I	I	C	A
1	iid_stable_null	8,000	P	I	I	I	I	I	I	C	A
2	iid_near_tie_null	8,000	P	C	I	I	I	I	I	C	A
3	exact_ties_null	2,000	P	E	E	E	E	E	E	E	A
4	practical_boundary	8,000	M	C	I	I	I	I	I	C	A
5	small_sign_reversal	8,000	M	C	I	I	I	I	I	C	A
6	medium_reversal	8,000	M	C	C	I	I	I	I	C	A
7	large_reversal	8,000	M	I	I	I	I	I	I	C	A
8	block_null	8,000	M	C	C	C	C	I	I	C	A
9	block_reversal	8,000	M	C	C	C	C	I	I	C	A
10	ar_null	8,000	M	C	C	C	C	C	I	C	A
11	ar_reversal	8,000	M	C	C	C	C	C	I	C	A
12	unequal_variances	8,000	M	I	I	I	I	I	I	C	A
13	mixed_distributions	8,000	M	I	I	I	I	I	I	C	A
14	mcar_reversal	8,000	M	I	I	I	I	I	I	C	A
15	selective_null	8,000	M	C	C	C	C	I	C	C	A
16	bounded_reversal	8,000	P	I	I	I	I	I	I	C	E
17	bounded_reversal_n400	4,000	P	I	I	I	I	I	I	I	E

A.4 Numerical boundary case and substantive limitations

The saved scenario 11, replicate 6628, comparison index 47 has a screened family of 43 entries. At rank 43 the stored p-value is 0.05. The original port evaluates $0.05 \times 43/43$ as 0.049999999999999996 and does not reject; the actual source evaluates $(43/43) \times 0.05$ as 0.05 and rejects. The mathematical bootstrap-grid comparison is $1/20 \leq 1/20$ and rejects. The old referee still reproduces its mismatch on this preserved record; the recovery contracts explicitly distinguish all three identities.

The reuse audit found 31 port-versus-source differences among the original 112,275 studies. At final aggregation, the largest difference in endpoint means between those implementations is 0.00025. Only the exact screened-BH addition changes final endpoint means relative to its original counterpart, with maximum change 8.56×10^{-7}; the other seven exact variants' endpoint means match their originals. Aggregate equality does not establish identity of every rejection vector.

These small arithmetic differences do not repair the larger statistical problems. Full-family historical BH has practical FDR 0.51875 in the block null and 0.778625 in the AR null. Grouped-bootstrap BY has ordinary coverage 0.88775 in the IID equal null and 0.759875 in the AR null. Under AR, cluster-t coverage is 0.8600 in the null and 0.864375 in the reversal scenario because fixed five-window clusters remain dependent. Conversely, low FDR can coexist with negligible power: in the block-reversal scenario, cluster-t practical power is approximately 0.00000658, oracle-normal 0.000296, and grouped-bootstrap zero. Bounded Hoeffding power is zero at $n = 40$ and 0.052632 at $n = 400$, with original-family bounds approximately $[0, 0.00401]$ and $[0.04462, 0.06065]$. Inconclusiveness, abstention, and conservatism remain findings rather than reasons to select a new reporting threshold.

B Evidence Identities and Reproduction Scope

The C01 full run is `c01-full-167b9074e5711420`; the recovery run is `stat01-recovery-v2-061933a64e27da20`. Execution commits are `2d8342e699f35044044ac518c0b75a3c4a1c768c` and `1ac4c8dd1421aabad05e6ce243cb911a6ed6c8e1`, respectively. The evidence review is pinned at commit `6b2eedd7aa254986b85c1bb901587641cbeac8b0`; the prewrite audit at `d22c7751af9c563d5a4d8a304c007f26112c4d12`. The canonical frozen-window hash begins `0d1aa06ccbee`; the physical window-file hash begins `97fbaf6a4b9b`; absolute C01 metrics begin `5700c4bbd738`. Full hashes, source paths, and stored-copy hashes are supplied in `evidence/input-hashes.json` and the input manifest.

The completed C01 campaign performed exhaustive event verification before publishing its completion marker. The final review separately checked all 5,760 terminal records and recomputed their metrics, all 164 launch-file fingerprints, all 144 reused identities, and all 720 request-condition records. Its fresh event replay covered 24 new-cell journals, structurally chosen from the first nonpilot window per source at HIGH_PRESSURE, both capacities, and all adaptations. Independent report checking covers 5,760 absolute rows, 15,840 rankings, 95,040 matched differences, and 26,136 reversal groups. M3 regenerates displays only from those saved summaries and verifies their fingerprints.

STAT01's recovery validated all 5,360 batches during execution. The final review checked the complete batch inventory and independently recombined all final endpoint moments with high-precision arithmetic; its fresh record-level check covered 200 saved studies across eight scenarios, including the original failing record. It did not claim to rerun all raw studies. Recovery preserves original bytes and labels post-diagnosis comparisons separately. Full raw-record validation and display regeneration are different reproducibility tasks; this manuscript package supplies the latter without generating any new experimental outcome.

The historical evidence remains under `paper/tompecs-m2`, including its statistical audit, claim changes, cross-metric set comparisons, and finite-reference ranking-recovery analyses. Those materials remain archived support rather than additional independent experiments. Related headroom and module-intervention projects share some infrastructure and topics, but their causal or native-engine results are not used as evidence for this paper. The author-review checklist requests current publication status, precise reuse disclosure, release permissions, and reviewer access before submission.